A Project Report

on

GENERATIVE AI CAREER GUIDANCE ASSISTANT

submitted in partial fulfilment of the requirement for the award of the degree of

Bachelor of Technology

in

Computer Science and Engineering

by

Akshit Sashi Kumar 230BTCCSE041

Deepak 230BTCCSE055

Utsav Kumar 230BTCCSE121

Under the Supervision of

Dr. Meenakshi Gupta (Assistant Professor, SET)

SCHOOL OF ENGINEERING AND TECHNOLOGY

SUSHANT UNIVERSITY, GURUGRAM-122011

JAN-MAY 2026

TABLE OF CONTENTS

Page No.

Declaration
Acknowledgement
Abstract
List of Tables
List of Figures

1. Introduction

1.1 Motivation
1.2 Problem Identification

2. Literature Review
2.1. Related works
2.2. Research Gaps
2.3. Objectives

3. Design/Modelling
3.1. Working principle
3.1.1 Block Diagram/Model/Flow chart/UCD/DFD
3.1.2 Circuit Diagram/Layout
3.2. Solution Methodology
3.3. List & specifications of components/material/Software/tools used

3. Results & Discussions
3.1 Findings/Outcome
3.1.1 Sub - Subsection
3.2 Subsection

5. Conclusions & Future Scope

Appendix
References
Gantt Chart
Responsibility Matrix
Similarity Score/Plag report

Declaration

I/We,Akshit Sashi Kumar (230BTCCSE041), Deepak (230BTCCSE055), and Utsav Kumar (230BTCCSE121), hereby declare that the project work titled **Generative AI Career Guidance Assistant** is a result of my/our original work and has not been submitted elsewhere for the award of any degree or diploma. All sources of information and material that have been consulted during the preparation of this project report have been duly acknowledged. I/we further declare that the work contained in this report is free from any form of plagiarism.

I/we understand that any act of academic dishonesty in the preparation of this report will lead to the disqualification of my/our project and may result in disciplinary action as deemed appropriate by the institution.

Date:

Signature:

Akshit Sashi Kumar
(230BTCCSE041)

Deepak

(230BTCCSE055)

Utsav Kumar

(230BTCCSE121)

Acknowledgment

I would like to express my sincere gratitude to all those who have contributed to the successful completion of this project.**Generative AI Career Guidance Assistant.**

I extend my heartfelt thanks to my project supervisor, **Dr. Meenakshi Gupta**, for their invaluable guidance, support, and encouragement throughout the project duration. Their expertise and insights have been instrumental in shaping the direction and quality of this work.

I am thankful to the faculty members of the School of Engineering and Technology, Sushant University for providing the necessary infrastructure and resources for the successful completion of this project.

Last but not least, I express my gratitude to my family and friends for their unwavering support and encouragement.

Submitted By:

- **Akshit Sashi Kumar** (230BTCCSE041)
- **Deepak** (230BTCCSE055)
- **Utsav Kumar** (230BTCCSE121)

Submitted To:

Dr. Meenakshi Gupta
School of Engineering and Technology

ABSTRACT

The **Generative AI Career Guidance Assistant** is a web-based intelligent system designed to support students, particularly those pursuing Computer Science and Engineering, in making structured and informed career decisions. In modern academic environments, students frequently encounter confusion while selecting career paths, identifying required skills, planning projects, and preparing for internships or placements. These challenges arise due to fragmented guidance, lack of continuity in mentoring, and absence of a centralized advisory system.

This project addresses these issues by developing an AI-powered conversational assistant that provides personalized, context-aware career guidance. Unlike traditional systems that rely on static recommendations or one-time assessments, the proposed solution enables continuous interaction through chat-based sessions, allowing students to revisit previous discussions and refine their career plans over time.

The system is implemented using a modern technology stack comprising:

- **Frontend:** React with Vite for dynamic and responsive user interface
- **Backend:** FastAPI for efficient API handling and asynchronous processing
- **Database:** SQLite for lightweight and structured data storage
- **AI Component:** Generative AI model for contextual response generation

The core functionality of the system revolves around three key components:

1. **User Profile Management –** Stores academic background, skills, and career interests
2. **Session-Based Chat System –** Allows multiple counselling discussions across topics
3. **Contextual AI Responses –** Generates personalized suggestions based on stored data

The development of the project was carried out in multiple phases, as documented in the fortnight reports. In the initial phase, the focus was on understanding the problem domain, defining system architecture, and setting up the frontend and backend structure . The second phase involved implementing core backend functionalities, integrating SQLite for persistent storage, and developing the chat interface with multi-session support . In the final phase, the system was refined with enhanced personalization, improved user interface design, structured response formatting, and complete end-to-end testing to ensure system reliability .

Further improvements were explored in extended development stages, including features such as resume-based skill extraction, enhanced AI response quality, and better usability through modern UI enhancements . These enhancements contribute to making the system more practical and user-friendly for real-world applications.

The results of the project demonstrate that integrating Generative AI with user context and session managementsignificantly improves the quality and relevance of career guidance. The system successfully enables:

- Continuous and structured career discussions
- Personalized recommendations based on user data
- Efficient storage and retrieval of past conversations
- Improved user engagement through an interactive interface

However, the current implementation is a functional prototype, and certain limitations remain, such as dependency on predefined AI behavior, lack of large-scale user testing, and absence of deployment in a real academic environment.

In conclusion, the project highlights the potential of Generative AI in transforming traditional career counselling into a scalable, accessible, and intelligent digital solution. Future enhancements may include integration with live job portals, advanced resume analysis, voice-based interaction, and improved AI model tuning for higher accuracy and reliability.

LIST OF TABLES

Table No.	Title	Page No.
Table 1.1	Problem Identification and Proposed Project Response	1
Table 2.1	Summary of Literature Insights and Identified Research Gaps	2
Table 3.1	Software, Frameworks, and Tools Used in the System	4
Table 4.1	Functional Testing and Validation Summary	5
Table A.1	Fortnight-wise Project Progress Log	7
Table G.1	Gantt Chart for Project Development Timeline	9
Table R.1	Responsibility Assignment Matrix (RACI Model)	10

The List of Tables section provides a structured overview of all tabular data used throughout the report. Tables play an essential role in presenting complex information in a concise, organized, and easily interpretable format. In this project, tables are strategically used to summarize problem analysis, research insights, system components, testing results, and project management details.

Table 1.1: Problem Identification and Proposed Project Response

This table highlights the key challenges faced by students in career planning and maps them to the corresponding solutions provided by the system. It clearly demonstrates how the project addresses real-world issues such as lack of personalized guidance, absence of continuous counselling, and poor tracking of career discussions.

Table 2.1: Literature Insights and Research Gaps

This table summarizes findings from existing systems such as rule-based career tools, counselling systems, and AI-based assistants. It also identifies critical research gaps like:

- Lack of session-based interaction
- Absence of personalized context handling
- Poor integration of profile and conversation data

These gaps directly influenced the design of the proposed system.

Table 3.1: Software and Tools Used

This table lists all technologies used in the project along with their purpose and justification. For example:

- React for UI development
- FastAPI for backend services
- SQLite for database management
- Generative AI for intelligent responses

It reflects the technical decisions made during system development.

Table 4.1: Functional Testing Summary

This table presents the testing results of different system modules, including:

- User profile handling
- Session management
- Chat rendering
- Data persistence

It validates that the system works as expected and meets functional requirements.

Table A.1: Fortnight-wise Progress Log

This table is derived from the project's fortnight reports and shows phase-wise development progress, including:

- Initial planning and architecture design
- Backend and database implementation
- UI improvements and testing

As seen in the progress documents, the project followed a step-by-step incremental development approach .

Table G.1: Gantt Chart

This table represents the timeline of project activities, showing how tasks such as research, development, testing, and documentation were distributed over time. It ensures proper planning and tracking of project milestones.

Table R.1: Responsibility Matrix

This table defines the roles and responsibilities of team members using the RACI model:

- Responsible (R)
- Accountable (A)
- Consulted (C)
- Informed (I)

LIST OF FIGURES

Figure No.	Title	Page No.
Fig. 3.1	Workflow of the Generative AI Career Guidance Assistant	3
Fig. 3.2	High-Level System Architecture	3
Fig. 3.3	Data Flow and Session Management Layout	4
Fig. 4.1	Implemented Feature Outcome Map	5
Fig. 5.1	Future Enhancement Roadmap	6

Detailed Explanation of Figures (For Report Expansion)

The List of Figures section provides a structured reference to all diagrams, flowcharts, and visual representations used in the report. Figures are essential for explaining system design, workflow, and architecture in a simplified and visually understandable manner. They complement textual explanations and help readers grasp complex concepts more efficiently.

Fig. 3.1: Workflow of the Generative AI Career Guidance Assistant

This figure represents the complete working flow of the system, starting from user input to AI-generated output. It includes:

- User interaction through the frontend interface
- Request processing via backend APIs
- Context extraction using user profile and session data
- Response generation using the AI model
- Storage of messages in the database

This workflow clearly shows how different components interact to deliver a seamless counselling experience.

Fig. 3.2: High-Level System Architecture

This diagram illustrates the three-layer architecture of the system:

1. **Presentation Layer –** Handles user interface (React + Vite)
2. **Application Layer –** Manages backend logic (FastAPI)
3. **Data & AI Layer –** Stores data (SQLite) and generates responses

The separation of layers ensures modularity, scalability, and easier maintenance of the system.

Fig. 3.3: Data Flow and Session Management Layout

This figure focuses on how data moves within the system, especially during chat interactions. It highlights:

- Message flow from frontend to backend
- Session identification and tracking
- Storage and retrieval of chat history
- Context building for AI responses

It is particularly important because the system relies on persistent conversations rather than one-time interactions.

Fig. 4.1: Implemented Feature Outcome Map

This figure summarizes the overall development process and achieved outcomes, divided into stages:

- Planning
- Implementation
- Refinement
- Testing

It visually demonstrates how the project evolved from an idea into a functional prototype.

Fig. 5.1: Future Enhancement Roadmap

This diagram presents the planned future improvements of the system, including:

- Resume analysis
- Job portal integration
- Voice-based interaction
- Deployment strategy

It provides a clear direction for extending the project beyond its current scope.

Role of Figures in This Project

Figures play a critical role in this report by:

- Simplifying complex technical workflows
- Providing visual clarity for architecture and design
- Supporting explanations with graphical representation
- Improving overall readability and presentation quality

In technical projects like this, diagrams are especially useful for explaining system interactions, data flow, and component relationships, which may be difficult to understand through text alone.

CHAPTER-1

INTRODUCTION

In the modern digital era, the rapid advancement of technology has significantly transformed the way students plan and build their careers. Particularly in the field of Computer Science and Engineering (CSE), students are exposed to a wide range of domains such as Artificial Intelligence, Data Science, Web Development, Cybersecurity, and Software Engineering. While this diversity provides numerous opportunities, it also creates confusion and uncertainty among students regarding which career path to choose and how to prepare for it effectively.

Traditionally, career guidance has been provided through faculty mentoring, seminars, workshops, and counselling sessions. Although these methods are valuable, they often suffer from several limitations such as:

- Limited availability of mentors
- Lack of personalized guidance
- Absence of continuous tracking of student progress
- Fragmented and unstructured advice

As a result, students frequently rely on scattered sources such as online articles, social media, or peer discussions, which may not always provide accurate or relevant guidance.

To address these challenges, this project introduces a Generative AI-based Career Guidance Assistant, a web-based system designed to provide interactive, personalized, and continuous career counselling support. The system uses conversational AI techniques to allow students to ask questions in natural language and receive structured guidance based on their individual profiles.

Background and Context

With the emergence of Generative AI and Large Language Models (LLMs), it has become possible to develop intelligent systems capable of understanding user queries and generating meaningful, context-aware responses. These technologies have already been applied in domains such as education, healthcare, and customer support, demonstrating their ability to improve accessibility and efficiency.

In the context of career guidance, Generative AI offers several advantages:

- It enables real-time interaction with users
- It supports natural language communication
- It can generate customized recommendations based on user data
- It allows continuous learning and improvement

However, most existing AI-based tools provide generic responses because they do not utilize structured user information or maintain conversation history. This project overcomes this limitation by integrating user profiles, session management, and message storage, enabling the system to deliver more relevant and personalized guidance.

Overview of the Proposed System

The proposed system is designed as a full-stack web application that combines frontend, backend, database, and AI components into a unified platform. The key idea is to create a system where students can:

- Maintain their academic and career profile
- Initiate multiple counselling sessions based on different topics
- Ask career-related questions in a conversational format
- Receive structured and contextual responses
- Revisit past discussions for better decision-making

The system ensures that each interaction is not treated as an isolated query but as part of a continuous career development journey.

The development of this system was carried out in multiple phases, as highlighted in the fortnight progress reports. Initially, the focus was on understanding the problem statement, defining system architecture, and setting up frontend and backend environments . Later phases involved implementing backend functionalities, database integration, and chat system development, followed by refinement, testing, and personalization improvements .

Need for the System

The need for a system like the Generative AI Career Guidance Assistant arises due to the following reasons:

1. **Lack of Personalized Guidance**
   Most students receive generalized advice that does not consider their individual skills, interests, and goals.
2. **Unavailability of Continuous Support**
   Career counselling is often limited to specific sessions, leaving students without immediate guidance when needed.
3. **Absence of Structured Planning**
   Students may know their goals but lack a clear roadmap to achieve them.
4. **Loss of Previous Discussions**
   Traditional counselling does not maintain records of past interactions, making it difficult to track progress.
5. **Increasing Complexity of Career Choices**
   With the rapid growth of technology fields, students face difficulty in choosing the right domain.

Significance of the Project

This project is significant because it demonstrates how Generative AI can be effectively applied in the domain of education and career development. It provides a scalable and accessible solution that can support a large number of students simultaneously.

Key contributions of the project include:

- Development of a context-aware conversational system
- Integration of profile-based personalization
- Implementation of session-based interaction model
- Creation of a persistent and structured guidance system

1.1 Motivation

The motivation behind developing the **Generative AI Career Guidance Assistant** arises from real challenges observed among students pursuing technical education, especially in the field of Computer Science and Engineering. During the academic journey, students are required to make several critical decisions that directly influence their future career paths. These decisions include selecting the right programming languages, choosing suitable projects, preparing for internships, and identifying long-term career goals such as higher education or job placement.

However, it has been observed that students often face uncertainty, confusion, and lack of direction due to the following reasons:

1. Increasing Complexity of Career Options

With the rapid advancement of technology, new domains such as:

- Artificial Intelligence
- Machine Learning
- Data Science
- Cloud Computing
- Cybersecurity
- Web and App Development

have emerged as major career opportunities. While this diversity is beneficial, it also creates a situation where students struggle to decide:

- Which domain aligns with their interests
- What skills are required for each domain
- How to start and progress in a chosen field

This complexity highlights the need for a system that can simplify decision-making through structured and guided suggestions.

2. Lack of Personalized Guidance

Traditional career guidance methods often provide general advice that is not tailored to individual students. For example:

- Two students with different skill levels may receive the same recommendation
- Career suggestions may not consider personal interests or academic background
- Guidance may not evolve as the student progresses

The motivation for this project is to create a system that uses user-specific data (profile, skills, goals) to generate personalized recommendations, making the guidance more relevant and actionable.

3. Limited Accessibility to Mentors

Although faculty members and career counsellors play a crucial role in guiding students, their availability is often limited due to:

- High student-to-mentor ratio
- Time constraints
- Scheduled counselling sessions

As a result, students may have to wait for guidance or proceed without proper direction. This project aims to provide an always-available digital assistant that can act as a first-level support system for students.

4. Lack of Continuity in Career Discussions

One of the major issues in traditional counselling is the absence of continuous interaction. Typically:

- Discussions are not recorded
- Previous advice is not easily accessible
- Students may forget earlier recommendations

This leads to repeated questions and inconsistent planning. The proposed system solves this by maintaining:

- Chat history
- Session-based discussions
- Persistent storage of messages

This ensures that career guidance becomes a continuous and evolving process rather than a one-time interaction.

5. Need for Structured Career Planning

Many students have a general idea of their goals but lack a clear roadmap to achieve them. For example:

- A student may want to become a data scientist but does not know the step-by-step process
- Another student may want a software job but lacks clarity on required projects and skills

This project is motivated by the need to provide step-by-step guidance, including:

- Skill development
- Project recommendations
- Internship preparation
- Resume building

6. Inspiration from Fortnight Development Phases

The motivation also evolved during the development phases documented in the project progress reports. Initially, the focus was on solving the problem of scattered and unstructured career guidance by designing a unified system . As the project progressed, the importance of contextual responses and session management became evident, leading to further refinement of the system .

In later stages, improvements such as better user interface design, personalized response formatting, and enhanced usability were incorporated, strengthening the system's ability to deliver meaningful guidance .

7. Role of Generative AI in Motivation

The emergence of Generative AI technologies provided a strong foundation for this project. Unlike traditional systems, Generative AI can:

- Understand natural language queries
- Generate human-like responses
- Adapt answers based on context
- Provide detailed explanations and suggestions

This capability inspired the development of a system where students can interact naturally, just like they would with a human mentor.

1.2 Problem Identification

In the current academic environment, students face multiple challenges while planning their careers. Despite the availability of information through the internet, educational institutions, and peer networks, there is still a significant gap between available

guidance and effective decision-making. This section identifies the major problems that motivated the development of the **Generative AI Career Guidance Assistant**.

1. Key Issues in Existing Career Guidance Systems

1.1 Generalized Advice

Most existing career guidance platforms provide generic recommendations that are not tailored to individual students. For example:

- All students interested in programming may be advised to learn the same technologies
- Suggestions often ignore the student's current skill level or background
- Career paths are recommended without considering personal interests

Impact:
Students may follow unsuitable paths, leading to confusion, loss of interest, and inefficient learning.

1.2 Lack of Continuous Interaction

Traditional counselling methods are often one-time or infrequent interactions. Students typically:

- Attend a single counselling session
- Receive advice without follow-up
- Lack support during implementation of suggestions

Impact:
Students struggle to maintain consistency in their career planning and may lose direction over time.

1.3 No Record of Previous Discussions

In most cases, career discussions are not recorded or stored systematically. This results in:

- Loss of important guidance
- Repetition of the same queries
- Difficulty in tracking progress

Impact:
Students are unable to build upon previous advice, making career planning inefficient.

1.4 Delayed Access to Guidance

Students often need immediate clarification for small doubts such as:

- Which project to choose next
- Whether a skill is relevant
- How to prepare for an internship

However, due to limited availability of mentors, students may not receive timely support.

Impact:
Delays in decision-making can slow down learning and preparation.

1.5 Lack of Structured Career Roadmap

Many students have goals but lack clarity on:

- Step-by-step skill development
- Required projects
- Industry expectations
- Preparation strategies

Impact:
Students may take random steps without a clear direction, reducing their chances of success.

2. Problem Summary Table

To clearly represent the identified problems and their corresponding solutions, the following table is presented:

Table 1.1: Problem Identification and Proposed Project Response

Issue Observed	Impact on Student	Proposed System Response
Advice is too general	Students receive the same suggestions regardless of their skills or goals	Use profile-based personalization for tailored guidance
Counselling is not continuous	Students lose direction between sessions	Provide an always-available chat-based assistant
No record of discussions	Previous guidance is lost	Store sessions and messages in the database
Delay in receiving help	Students cannot resolve doubts immediately	Enable real-time conversational interaction
Lack of structured planning	Students are unsure about next steps	Provide step-by-step career guidance

3. Insights from Project Development Phases

The problems identified above were not only based on theoretical understanding but also emerged during the project's development process. According to the fortnight reports:

- In the initial phase, the need for a centralized guidance system was identified while analyzing student challenges
- In the implementation phase, issues related to context handling and session management were recognized and addressed
- In the refinement phase, the importance of user experience, personalization, and response quality became evident

These observations helped in shaping a system that directly addresses real-world problems.

4. Problem Statement

Based on the above analysis, the problem can be formally stated as:

> "There is a lack of an integrated, intelligent, and personalized system that can provide continuous career guidance to students by maintaining their profile, tracking their interactions, and generating context-aware recommendations."

5. Scope of the Proposed Solution

To address the identified problems, the project focuses on developing a system with the following capabilities:

- Maintain student profiles for personalization
- Support multiple chat sessions for different career topics
- Store and retrieve conversation history
- Generate context-aware AI responses

CHAPTER- 2

LITERATURE REVIEW

The Literature Review provides an overview of existing research, tools, and systems related to career guidance, recommendation systems, and conversational AI. It helps in understanding the current state of technology and identifying gaps that the proposed system aims to address.

In recent years, the integration of Artificial Intelligence (AI) and Machine Learning (ML) in educational and career counselling systems has gained significant attention. Various approaches have been proposed, ranging from rule-based systems to advanced AI-driven conversational assistants.

2.1 Related Works

This section discusses the major categories of existing systems relevant to the project and analyzes their strengths and limitations.

1. Rule-Based Career Guidance Systems

Rule-based systems are among the earliest approaches used for career recommendation. These systems typically rely on:

- Predefined rules
- Questionnaires or aptitude tests
- Fixed decision trees

Based on user responses, the system suggests suitable career options.

Advantages:

- Easy to design and implement
- Predictable and controlled outputs
- Useful for basic guidance

Limitations:

- Cannot handle complex or open-ended queries
- Lack flexibility and adaptability
- Provide limited personalization

Relevance to Project:
These systems highlighted the importance of structured input but also demonstrated the need for a more flexible and intelligent approach.

2. Counsellor-Based Guidance Systems

Traditional career guidance often involves human counsellors who provide advice based on:

- Student interaction
- Academic performance
- Personal interests

These systems are widely used in educational institutions.

Advantages:

- Highly personalized guidance
- Ability to understand emotional and contextual factors
- Reliable decision-making

Limitations:

- Limited availability of counsellors
- Time-consuming process
- Lack of scalability

Relevance to Project:
This approach inspired the need to replicate personalized interaction using an AI-based system while overcoming availability limitations.

3. Student Record Management Systems

These systems focus on storing and managing student information such as:

- Academic records
- Personal details
- Counselling notes

They are primarily used for administrative purposes.

Advantages:

- Organized data storage
- Easy retrieval of student information
- Useful for tracking progress

Limitations:

- No intelligent recommendation capability
- Lack of interactive features
- Limited role in decision-making

Relevance to Project:
This category influenced the inclusion of profile storage and session management in the proposed system.

4. Conversational AI Systems

With the rise of AI, conversational systems such as chatbots have become popular. These systems allow users to interact using natural language.

Features:

- Natural language understanding
- Interactive communication
- Automated response generation

Advantages:

- User-friendly interface
- Real-time interaction
- Scalable solution

Limitations:

- Often provide generic responses
- Lack deep personalization without context
- May not retain long-term conversation history

Relevance to Project:
This approach forms the core interaction model of the proposed system, enhanced with context-awareness and session storage.

5. Generative AI and LLM-Based Systems

Recent advancements in Generative AI and Large Language Models (LLMs) have revolutionized how systems generate responses. These models can:

- Understand complex queries
- Generate human-like responses
- Provide detailed explanations

Advantages:

- Highly flexible and adaptive
- Capable of handling open-ended questions
- Supports dynamic content generation

Limitations:

- Requires proper context for accurate responses
- Risk of generating irrelevant or generalized answers
- Needs careful prompt design

Relevance to Project:
The proposed system uses Generative AI to provide contextual and personalized career guidance, addressing the limitations of earlier systems.

6. Insights from Project Development

The understanding of these systems was further refined during the project development phases:

- Early stages focused on selecting an approach that balances flexibility and structure
- Implementation phases highlighted the importance of **c**ontext-aware responses and data persistence
- Final refinements emphasized improving response quality and user interaction experience

7. Summary Table of Related Works

Table 2.1: Summary of Literature Insights

Area	Observation	Relevance to Project
Rule-Based Systems	Fixed logic and limited flexibility	Highlighted need for dynamic AI-based system
Counsellor-Based Systems	Personalized but not scalable	Inspired conversational interaction
Record Systems	Organized data storage	Led to profile and session management
Conversational AI	Interactive but generic	Improved with context-based responses
Generative AI	Flexible and powerful	Used as core technology

Conclusion (Page 1 of Literature Review)

The study of existing systems reveals that while each approach has its strengths, none of them fully satisfies the need for a personalized, continuous, and intelligent career guidance system. This gap forms the foundation for the proposed solution, which combines the strengths of multiple approaches into a unified system.

2.2 Research Gaps

After analyzing existing career guidance systems, conversational AI tools, and recommendation approaches, several limitations and gaps have been identified. These gaps highlight why a new system is required and how the proposed **Generative AI Career Guidance Assistant** aims to improve upon existing solutions.

1. Lack of Personalization in Existing Systems

Most available systems either:

- Provide generic recommendations
- Use limited input parameters (e.g., interest or aptitude only)

They do not consider a comprehensive student profile, including:

- Academic background
- Skill level
- Career goals
- Previous interactions

Gap Identified:
Absence of deep personalization in guidance systems.

Proposed Solution:
The project integrates profile-based context handling, ensuring that responses are tailored to individual students.

2. Absence of Continuous Interaction

Many systems provide one-time recommendations without allowing users to:

- Ask follow-up questions
- Refine their career plans
- Continue discussions over time

Gap Identified:
Lack of continuous and evolving guidance.

Proposed Solution:
The system supports session-based conversations, enabling ongoing interaction and progressive career planning.

3. No Integration of Profile, Chat, and Planning

Existing tools often treat components separately:

- Profile data is stored but not used effectively
- Chat systems operate without context
- Planning tools are isolated

Gap Identified:
Lack of integration between key components.

Proposed Solution:
The project combines:

- User profiles
- Chat sessions
- Message history

into a unified workflow, ensuring consistency and relevance.

4. Limited Context Awareness in AI Systems

Although conversational AI systems are widely used, many of them:

- Do not retain previous interactions
- Generate responses without context
- Provide repetitive or generic answers

Gap Identified:
Weak contextual understanding in AI-based systems.

Proposed Solution:
The system uses:

- Stored profile data
- Session history
- Previous messages

to generate **c**ontext-aware responses.

5. Lack of Persistent Data Storage

Many AI tools do not store user interactions, leading to:

- Loss of previous discussions
- Repetition of queries
- Inefficient guidance process

Gap Identified:
No persistent storage of career guidance interactions.

Proposed Solution:
The project uses SQLite database to store:

- User profiles
- Chat sessions
- Messages

This ensures that all interactions are preserved and accessible.

6. Poor Scalability of Traditional Systems

Human counsellor-based systems are:

- Limited by availability
- Not scalable for large student populations
- Time-intensive

Gap Identified:
Inability to provide guidance to a large number of students simultaneously.

Proposed Solution:
The AI-based system provides a scalable solution, capable of handling multiple users at the same time.

7. Lack of Structured Career Roadmaps

Existing systems often fail to provide:

- Step-by-step guidance
- Clear learning paths
- Actionable recommendations

Gap Identified:
Absence of structured career planning.

Proposed Solution:
The system generates structured responses, including:

- Skills to learn
- Projects to build
- Career steps to follow

8. Insights from Development Phases

The research gaps were not only theoretical but also emerged during the project development:

- Early phases revealed the need for integrating multiple system components

- Implementation phases highlighted challenges in context handling and session management

- Final stages focused on improving personalization, usability, and response quality

These insights helped refine the system to address real-world challenges effectively.

9. Summary of Research Gaps

The identified gaps can be summarized as follows:

- Lack of personalization
- Absence of continuous interaction
- Poor integration of system components
- Weak context awareness in AI systems
- No persistent storage of interactions
- Limited scalability of traditional systems
- Lack of structured guidance

2.3 Objectives

The objectives of the **Generative AI Career Guidance Assistant** project define the goals and expected outcomes of the system. These objectives are formulated based on the problems identified and the research gaps observed in existing career guidance solutions. The aim is to develop a system that is not only technically functional but also practically useful for students.

1. Primary Objective

The primary objective of this project is:

> To design and develop a web-based Generative AI Career Guidance Assistant that provides personalized, context-aware, and continuous career guidance to students through an interactive conversational interface.

2. Specific Objectives

To achieve the primary goal, the project focuses on the following specific objectives:

2.1 Develop a User Profile Management System

- Create a system to store student details such as:
  - Academic background
  - Skills
  - Interests
  - Career goals
- Use this data to personalize AI-generated responses

Purpose:
To ensure that guidance is tailored to individual users rather than being generic.

2.2 Implement a Session-Based Chat System

- Allow users to create multiple chat sessions
- Enable topic-based discussions (e.g., skills, internships, projects)
- Maintain conversation history for each session

Purpose:
To support continuous interaction and organized career discussions.

2.3 Integrate Generative AI for Response Generation

- Use AI models to generate:
    - Career suggestions
    - Skill recommendations
    - Step-by-step guidance
- Ensure responses are:
    - Relevant
    - Structured
    - Context-aware

Purpose:
To provide intelligent and human-like career counselling.

2.4 Ensure Data Persistence and Retrieval

- Store:
    - User profiles
    - Chat sessions
    - Messages
- Enable retrieval of past interactions

Purpose:
To maintain continuity and improve user experience.

2.5 Design a User-Friendly Interface

- Develop a responsive frontend using React
- Ensure easy navigation and readability
- Implement features like:
    - Chat display
    - Session switching
    - Profile management

Purpose:
To enhance usability and encourage regular interaction.

2.6 Build a Scalable and Modular Architecture

- Separate system into:
  - Frontend
  - Backend
  - Database
- Use FastAPI for efficient backend operations

Purpose:
To allow future expansion and easy maintenance.

2.7 Perform Testing and Validation

- Test system functionality including:
  - Chat operations
  - Session handling
  - Data storage
- Validate system performance through:
  - End-to-end testing
  - User interaction scenarios

Purpose:
To ensure reliability and correctness of the system.

3. Objectives Based on Development Phases

The objectives evolved during the project development phases:

- Initial phase focused on system design and architecture planning
- Middle phase emphasized backend implementation and database integration
- Final phase aimed at improving personalization, UI design, and testing

This phased approach ensured that all objectives were systematically achieved.

4. Expected Outcomes

By achieving the above objectives, the system is expected to:

- Provide personalized career guidance
- Support continuous and interactive counselling
- Improve decision-making for students
- Offer a structured approach to career planning
- Demonstrate the practical application of Generative AI

CHAPTER-3

DESIGN AND METHODOLOGY

This chapter describes the overall design, architecture, and working methodology of the **Generative AI Career Guidance Assistant**. It explains how different components of the system interact with each other to provide personalized and continuous career guidance to users.

The system is designed using a modular and layered approach, ensuring that each component (frontend, backend, database, and AI module) functions independently while maintaining seamless communication with other components.

3.1 Working Principle

The working principle of the system is based on a profile-driven conversational workflow, where user inputs are processed through multiple stages to generate context-aware responses.

1. Overview of System Workflow

The system operates through the following sequence of steps:

1. **User Interaction (Frontend)**
    1. The user enters a query related to career guidance through the web interface.
    2. The interface is built using React and provides a chat-based interaction system.
2. **Request Handling (Backend API)**
    1. The user's query is sent to the backend using API calls.
    2. FastAPI processes the request and validates the session.
3. **Context Preparation**
    1. The system retrieves:
        1. User profile data
        2. Previous chat messages
        3. Active session details
    2. This information is used to build context for the AI model.
4. **AI Response Generation**
    1. The Generative AI model processes the query along with context.
    2. It generates a structured and personalized response.

5. **Data Storage (Database)**

    1. Both user queries and AI responses are stored in the SQLite database.
    2. This ensures persistence and future retrieval.

6. **Response Display (Frontend)**

    1. The generated response is sent back to the frontend.
    2. It is displayed in the chat interface in a readable format.

2. Key Features of Working Principle

The system is designed with the following important features:

2.1 Profile-Based Context Handling

- The system uses stored user data to understand:
    - Skill level
    - Interests
    - Career goals

Benefit:
Ensures that responses are personalized and relevant.

2.2 Session-Based Interaction

- Users can create multiple chat sessions for different topics.
- Each session maintains its own message history.

Benefit:
Allows organized and continuous discussions.

2.3 Persistent Data Storage

- All interactions are stored in the database.
- Users can revisit previous conversations anytime.

Benefit:
Prevents loss of important guidance and supports long-term planning.

2.4 Context-Aware AI Responses

- The AI model generates responses based on:
    - Current query
    - User profile
    - Previous messages

Benefit:
Improves the quality and relevance of guidance.

3. Conceptual Workflow Representation

The working principle can be conceptually understood as:

User → Frontend → Backend API → Context Processing → AI Model → Database → Frontend Display

This flow ensures smooth communication between all components and maintains system efficiency.

4. Insights from Development Phases

The working principle was refined during different stages of development:

- Initial phase focused on designing the basic workflow and system structure
- Implementation phase added database integration and session handling
- Final phase improved contextual response generation and UI interaction

These improvements ensured that the system evolved into a fully functional prototype.

5. Advantages of the Working Principle

The proposed working principle offers several advantages:

- Provides real-time career guidance
- Ensures personalized recommendations
- Maintains continuity in discussions
- Supports efficient data management
- Enhances user experience through interaction

3.1.1 Block Diagram / System Architecture

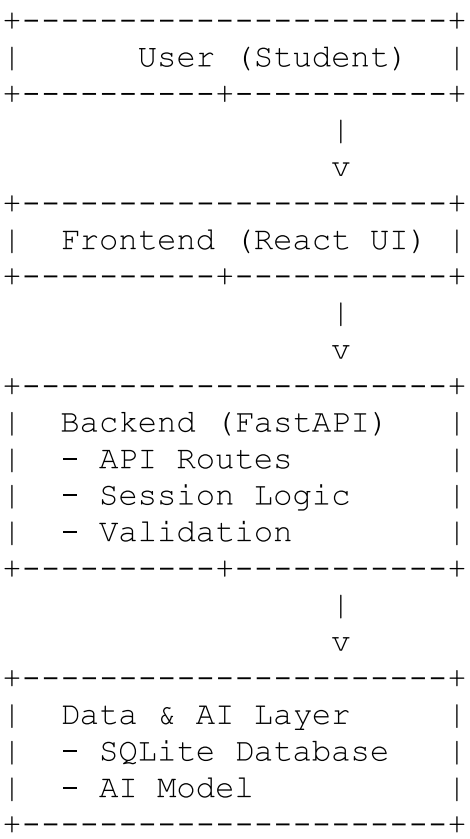
The system architecture of the Generative AI Career Guidance Assistant defines how different components of the system are organized and interact with each other. It follows a layered architecture model, which improves modularity, scalability, and maintainability.

The system is divided into three major layers:

1. Presentation Layer (Frontend)
2. Application Layer (Backend API)
3. Data and AI Layer (Database + AI Model)

1. Block Diagram Overview

The system can be represented using the following conceptual block diagram:



2. Description of Each Layer

2.1 Presentation Layer (Frontend)

The presentation layer is responsible for handling user interaction. It is developed using React with Vite, which allows efficient rendering and dynamic updates.

Key Components:

- Chat Interface
- Profile Management Forms
- Session Controls
- Navigation System

Functions:

- Accept user input (queries)
- Display AI-generated responses
- Manage session switching
- Provide a user-friendly interface

Advantages:

- Fast and responsive UI
- Reusable components
- Improved user experience

2.2 Application Layer (Backend)

The backend is developed using FastAPI, which handles all server-side operations and acts as a bridge between the frontend and data layer.

Key Responsibilities:

- API request handling
- Input validation
- Session management
- Context preparation
- Communication with AI model
- Database operations

Important Modules:

- User Management Module
- Profile Handling Module
- Chat Session Module
- Message Processing Module

Advantages:

- High performance and asynchronous processing
- Easy API development
- Scalable architecture

2.3 Data and AI Layer

This layer consists of:

a. SQLite Database

- Stores:
    - User profiles
    - Chat sessions
    - Messages
- Provides persistent storage

b. Generative AI Model

- Processes user queries
- Generates context-aware responses
- Provides career guidance

Advantages:

- Efficient data management
- Context-aware intelligence
- Lightweight and easy to maintain

3. Interaction Between Components

The interaction between layers follows a structured process:

1. User sends a query through the frontend
2. The request is sent to the backend API
3. Backend retrieves:
    - User profile
    - Session history
4. Context is prepared
5. AI model generates response
6. Response is stored in database
7. Response is displayed on frontend

This ensures smooth communication and data flow across the system.

4. Architectural Design Principles

The system is designed based on the following principles:

4.1 Modularity

Each component is independent and can be modified without affecting the entire system.

4.2 Scalability

The system can be extended by:

- Replacing SQLite with a larger database
- Integrating advanced AI models
- Adding new features

4.3 Maintainability

Clear separation of layers makes the system easier to debug and update.

4.4 Efficiency

FastAPI ensures quick request handling, while React ensures efficient UI updates.

5. Development Insights

The architecture evolved during the project development phases:

- Initial design focused on basic frontend-backend communication
- Later stages added database integration and session management
- Final improvements focused on enhancing performance, UI, and AI interaction

6. Advantages of the Proposed Architecture

- Supports real-time interaction
- Enables personalized guidance
- Ensures data persistence
- Allows future scalability and improvements
- Provides clear separation of concern

3.1.2 System Layout

The **system layout** defines how different functional components of the Generative AI Career Guidance Assistant are organized and how users interact with the system. It focuses on the arrangement of modules and their relationships within the application.

The layout is designed to ensure ease of use, logical flow of operations, and efficient data handling.

1. Major Components of System Layout

The system layout consists of the following key components:

1.1 User Profile Module

This module is responsible for storing and managing user-specific information such as:

- Academic details
- Technical skills
- Interests
- Career goals

Functionality:

- Create and update profile
- Provide data for personalization
- Support context-aware responses

Importance:
This module acts as the foundation for generating relevant and customized career guidance.

1.2 Chat Session Module

This module allows users to manage multiple counselling sessions.

Features:

- Create new sessions
- Switch between sessions
- Delete or manage sessions
- Maintain separate topics (e.g., internships, projects, skills)

Importance:
Enables organized and topic-wise career discussions, improving clarity and usability.

1.3 Message Management Module

This module handles all communication between the user and the AI system.

Functions:

- Store user queries
- Store AI responses
- Retrieve message history
- Display structured chat

Importance:
Ensures continuity of conversations and prevents loss of important information.

1.4 AI Response Module

This is the core intelligence layer of the system.

Functions:

- Process user input
- Use profile and session data as context
- Generate meaningful career guidance

Importance:
Provides intelligent and human-like interaction.

1.5 Database Module (SQLite)

The database stores all persistent data including:

- User profiles
- Sessions
- Messages

Importance:
Ensures data persistence, reliability, and retrieval.

2. Layout Structure (Conceptual View)

The system layout can be summarized as:

User → Profile Module → Chat Session Module → Message Module → AI Module → Database

This structured layout ensures that each component performs a specific role while contributing to the overall system functionality.

3. Data Flow in the System

The data flow describes how information moves through the system during a typical user interaction.

Step-by-Step Data Flow Process

1. **User Input**
    - The user enters a query in the chat interface.
2. **Frontend Processing**
    - The input is captured and sent to the backend via API.
3. **Backend Processing**
    - Backend validates:
        - User identity
        - Active session
    - Retrieves:
        - Profile data
        - Previous messages
4. **Context Formation**
    - The system combines:
        - Current query
        - User profile
        - Session history
5. **AI Processing**
    - The AI model generates a response based on the context.
6. **Data Storage**
    - Both query and response are stored in the database.
7. **Response Delivery**
    - The response is sent back to the frontend.

8. **Display to User**
    - The chat interface updates with the new response.

4. Data Flow Characteristics

4.1 Continuous Flow

The system maintains a continuous flow of information across sessions, enabling long-term interaction.

4.2 Context-Aware Processing

Data flow is not isolated; it incorporates:

- Past messages
- User profile

This ensures meaningful responses.

4.3 Bidirectional Communication

- User → System (Query)
- System → User (Response)

This creates an interactive experience.

5. Importance of System Layout and Data Flow

The system layout and data flow are crucial because they:

-
- Ensure smooth interaction between components
- Enable efficient data handling
- Support personalization and context-awareness
- Improve system performance and usability

6. Insights from Development Phases

The system layout evolved during the development process:

- Initial design focused on basic component structure and workflow planning
- Implementation phase added database connectivity and message handling
- Final phase improved data flow efficiency and UI interaction.

3.2 Solution Methodology

The solution methodology describes the step-by-step approach followed to design, develop, and implement the **Generative AI Career Guidance Assistant**. The development process was carried out in a structured and incremental manner, ensuring that each component of the system was properly designed, implemented, and tested.

The methodology follows a phased development approach, where the system is built progressively over multiple stages, as documented in the fortnight progress reports.

1. Overview of Development Approach

The project development is divided into three major phases:

1. Planning and Design Phase
2. Implementation Phase
3. Refinement and Testing Phase

Each phase contributes to building a complete and functional system.

2. Phase 1: Planning and Design

In the initial phase, the focus was on understanding the problem and designing the system architecture.

Activities Performed:

- Identification of project scope and objectives
- Study of existing systems and technologies
- Design of system workflow and architecture
- Selection of technology stack:
    - React (Frontend)
    - FastAPI (Backend)
    - SQLite (Database)
- Creation of basic UI structure
- Planning database schema for:
    - Users
    - Profiles
    - Sessions
    - Messages

Outcome:
A clear blueprint of the system was established, defining how different components would interact.

As observed in the initial fortnight report, this phase focused on system planning, architecture design, and environment setup

3. Phase 2: Implementation

This phase involved developing the core functionality of the system.

Activities Performed:

- Development of frontend components:
  - Chat interface
  - Profile forms
  - Navigation system
- Backend development using FastAPI:
  - API routes for user and session management
  - Request handling and validation
- Database integration:
  - SQLite setup
  - Table creation
  - Data storage and retrieval
- Implementation of chat system:
  - Message handling
  - Session-based interaction
- Integration of AI model for response generation

Outcome:
A working prototype was developed with basic functionality.

According to the development progress, this phase focused on backend implementation, database integration, and chat functionality

4. Phase 3: Refinement and Testing

In the final phase, the system was improved and validated.

Activities Performed:

- Enhancement of user interface:
  - Responsive design
  - Improved navigation
  - Dark theme implementation

- Improvement of AI responses:
    - Context-based generation
    - Structured output formatting
- Testing and validation:
    - Functional testing
    - End-to-end testing
    - Session handling verification
- Performance improvements:
    - Faster response handling
    - Efficient data retrieval

Outcome:
A refined and user-friendly system with improved performance and reliability.

As highlighted in the progress reports, this phase emphasized personalization, UI improvement, and system testing

5. Methodology Flow (Conceptual Representation)

The overall methodology can be summarized as:

Requirement Analysis → System Design → Implementation → Testing → Refinement

This iterative process ensures continuous improvement of the system.

6. Key Features of the Methodology

6.1 Incremental Development

- System developed in small stages
- Each phase builds upon the previous one

Benefit:
Reduces errors and improves system stability.

6.2 Modular Approach

- Separate development of frontend, backend, and database

Benefit:
Easier debugging and maintenance.

6.3 Continuous Testing

- Testing performed at each stage

Benefit:
Ensures system reliability and correctness.

6.4 User-Centric Design

- Focus on usability and interaction

Benefit:
Improves user experience and engagement.

7. Advantages of the Adopted Methodology

- Structured and organized development process
- Easy identification and resolution of issues
- Flexibility to incorporate changes
- Improved system quality and performance
- Efficient project management

3.3 Software, Tools and Technologies Used

The development of the **Generative AI Career Guidance Assistant** required the use of various software tools, frameworks, and technologies. These tools were carefully selected based on factors such as performance, scalability, ease of development, and compatibility with the project requirements.

This section provides a detailed overview of the technologies used along with their purpose and justification.

1. Summary Table of Tools and Technologies

Table 3.1: Software, Frameworks, and Tools Used

Component/Tool	Specification / Use	Reason for Selection
React	Frontend JavaScript library for UI development	Enables dynamic and reusable components
Vite	Development and build tool	Fast development and efficient bundling
FastAPI	Backend framework (Python)	High performance and easy API development
SQLite	Relational database	Lightweight and suitable for prototype
Generative AI Model	Response generation engine	Enables natural language interaction
Browser UI	Interface for user interaction	Provides accessibility and usability

2. Detailed Description of Each Tool

2.1 React (Frontend Framework)

React is a widely used JavaScript library for building user interfaces.

Key Features:

- Component-based architecture
- Virtual DOM for efficient updates
- Reusable UI components

Usage in Project:

- Chat interface development
- Profile management forms
- Navigation and session controls

Reason for Selection:

- Fast rendering
- Strong community support
- Ideal for dynamic applications

2.2 Vite (Build Tool)

Vite is a modern frontend build tool that provides fast development and optimized builds.

Key Features:

- Instant server start
- Hot module replacement (HMR)
- Optimized production build

Usage in Project:

- Frontend project setup
- Development environment

Reason for Selection:

- Faster than traditional tools
- Simple configuration
- Improves developer productivity

2.3 FastAPI (Backend Framework)

FastAPI is a high-performance Python framework used for building APIs.

Key Features:

- Asynchronous request handling
- Automatic API documentation
- Easy integration with Python libraries

Usage in Project:

- API development
- Request handling
- Session management
- Communication with database and AI model

Reason for Selection:

- High speed and efficiency
- Easy to use
- Suitable for scalable applications

2.4 SQLite (Database)

SQLite is a lightweight relational database used for storing application data.

Key Features:

- File-based storage
- No server required
- Simple and efficient

Usage in Project:

- Storing user profiles
- Managing chat sessions
- Storing messages

Reason for Selection:

- Lightweight and easy to integrate
- Suitable for prototype development
- Minimal setup required

2.5 Generative AI Model

The Generative AI model is the core component responsible for generating responses.

Key Features:

- Natural language understanding
- Context-aware response generation
- Ability to handle open-ended queries

Usage in Project:

- Career guidance suggestions
- Skill recommendations
- Step-by-step planning

Reason for Selection:

- Provides intelligent interaction
- Enhances user experience

2.6 Browser-Based User Interface

The system is accessed through a web browser.

Features:

- Responsive design
- Easy accessibility
- No installation required

Usage in Project:

- User interaction
- Chat display
- Profile management

Reason for Selection:

- Platform-independent
- Widely accessible
- User-friendly

3. Integration of Technologies

All the selected tools are integrated to form a complete system:

- React handles the frontend interface
- FastAPI manages backend operations
- SQLite stores all persistent data
- AI model generates intelligent responses

This integration ensures smooth communication and efficient system performance.

4. Advantages of Selected Technologies

-
- **Performance:** FastAPI ensures high-speed backend processing
- **Scalability:** Modular architecture supports future expansion
- **Ease of Development:** React and Vite simplify frontend development
- **Cost Efficiency:** SQLite eliminates the need for complex database setup
- **User Experience:** AI model provides interactive and engaging responses

5. Insights from Development Phases

The choice of tools was validated during the development process:

- Initial setup focused on selecting suitable technologies and creating project structure
- Implementation phase confirmed the effectiveness of FastAPI and SQLite integration
- Final phase improved frontend usability and response quality

CHAPTER-4

RESULTS & DISCUSSIONS

This chapter presents the outcomes achieved after the development and implementation of the **Generative AI Career Guidance Assistant**. It evaluates the system's functionality, performance, and effectiveness in addressing the identified problems.

The results are based on the working prototype developed during the project and the testing performed across different modules.

4.1 Findings / Outcome

The project successfully resulted in the development of a functional web-based prototype that integrates frontend, backend, database, and AI components into a unified system. The system demonstrates the practical implementation of Generative AI in the domain of career guidance

1. Overview of Achieved Results

The system achieved the following major outcomes:

1.1 Functional Chat-Based Career Guidance System

- Users can interact with the system using natural language
- The AI generates structured and meaningful responses
- Conversations are displayed in an organized chat format

Impact:
Provides an intuitive and user-friendly interface for career counselling.

1.2 Profile-Based Personalization

- User profiles are created and stored
- AI responses are influenced by:
    - Skills
    - Interests
    - Career goals

Impact:
Ensures that guidance is tailored to individual users instead of being generic.

1.3 Multi-Session Support

- Users can create multiple chat sessions
- Each session can represent a different topic:

- Skills
- Internships
- Projects

Impact:
Improves organization and clarity in career discussions.

1.4 Persistent Data Storage

- All user data is stored in SQLite database
- Includes:
  - Profiles
  - Sessions
  - Messages

Impact:
Enables retrieval of past discussions and ensures continuity.

1.5 Context-Aware Response Generation

- AI responses consider:
  - Current query
  - User profile
  - Previous messages

Impact:
Improves relevance and quality of guidance.

2. Development Outcome Stages

The project outcomes can be understood through different development stages:

2.1 Planning Stage

- Defined system architecture
- Designed workflow and database schema

As observed in early development, the focus was on planning and system structure design

2.2 Implementation Stage

- Developed frontend and backend
- Integrated database and chat system

This stage focused on core system functionality and integration

2.3 Refinement Stage

- Improved UI design
- Enhanced AI responses
- Performed testing

Final stage emphasized usability, personalization, and testing

3. Outcome Representation

The overall outcome of the system can be summarized as:

Planning → Implementation → Refinement → Testing → Functional Prototype

This progression shows how the system evolved from a concept into a working application.

4. Key Achievements

The major achievements of the project include:

- Successful integration of Generative AI with web technologies
- Development of a context-aware conversational system
- Implementation of session-based interaction model
- Creation of a persistent and structured guidance platform

5. Comparison with Initial Objectives

Objective	Achieved Status
Profile management	Successfully implemented
Session-based chat	Fully functional
AI-based guidance	Implemented with context-awareness
Data persistence	Achieved using SQLite
User-friendly interface	Improved through UI refinement

4.1.1 System Interface and Output Screens

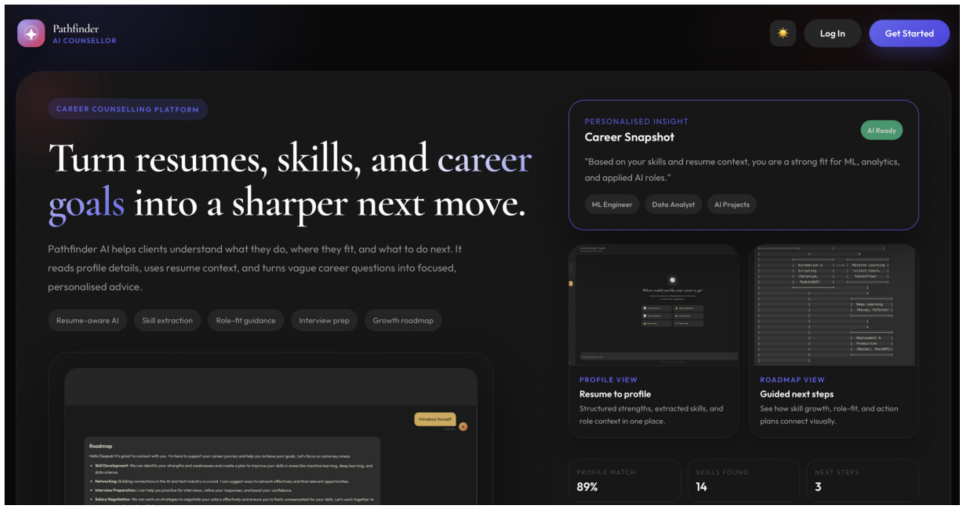


Fig. 4.1: Landing Page of the Generative AI Career Guidance Assistant

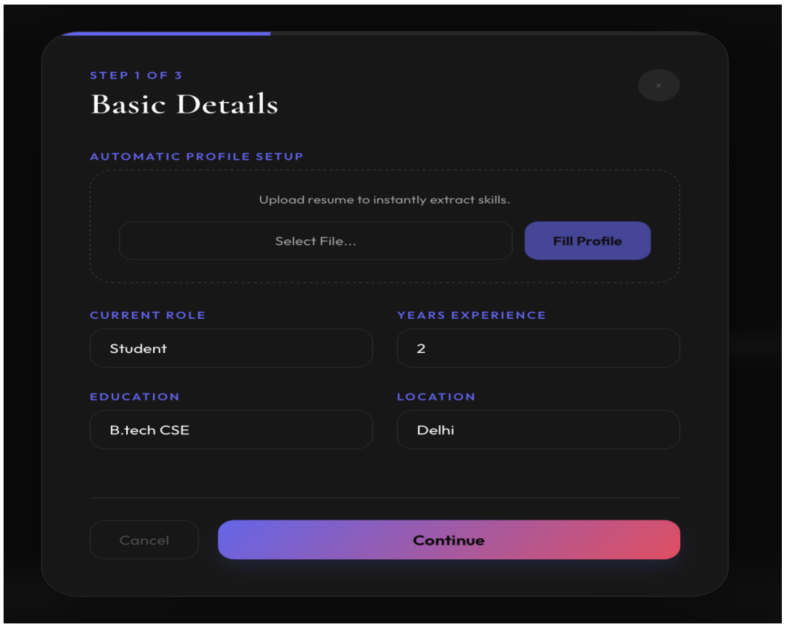


Fig 4.2 : User Profile Page where user store their skills and interest

Fig 4.3 : The Dashboard where we create new session and Chat session takes place

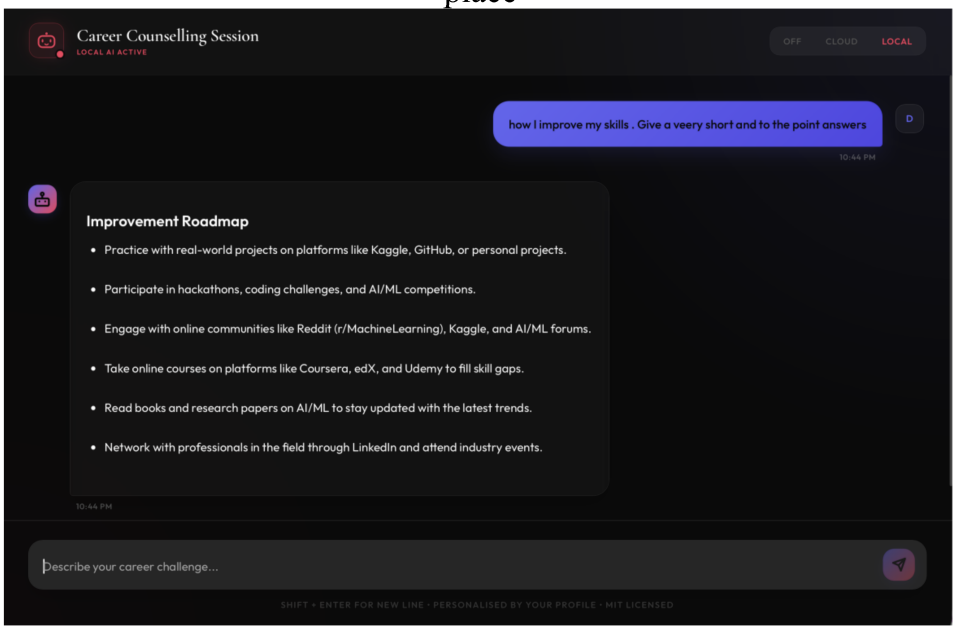


Fig 4.4 : The Chat with AI and Generating the response

4.2 Testing and Discussion

After the successful development of the **Generative AI Career Guidance Assistant**, extensive testing was carried out to evaluate the functionality, performance, and reliability of the system. This section discusses the testing process, observed results, and overall system behavior.

The testing phase ensures that the system meets the expected requirements and performs efficiently under different user scenarios.

1. Objectives of Testing

The main objectives of testing were:

- To verify that all system modules function correctly
- To ensure proper interaction between frontend, backend, and database
- To validate AI-generated responses
- To check system usability and user experience
- To identify and fix errors or inconsistencies

2. Types of Testing Performed

2.1 Functional Testing

Functional testing was performed to ensure that each feature of the system works as intended.

Tested Components:

- User registration and profile creation
- Chat session creation and switching
- Message sending and receiving
- AI response generation
- Data storage and retrieval

Result:
All major functionalities were successfully implemented and tested.

2.2 Integration Testing

Integration testing verified the interaction between different modules:

- Frontend ↔ Backend
- Backend ↔ Database
- Backend ↔ AI Model

Result:
Smooth communication was observed between all components, with no major integration issues.

2.3 Usability Testing

Usability testing focused on evaluating the user interface and experience.

Parameters Checked:

- Ease of navigation
- Readability of chat interface
- Responsiveness of UI
- User interaction flow

Result:
The system was found to be user-friendly, especially after UI improvements in later development stages.

2.4 Data Persistence Testing

This testing ensured that all user data is properly stored and retrieved.

Test Cases:

- Saving user profiles
- Storing chat sessions
- Retrieving previous messages

Result:
SQLite database successfully maintained all records, ensuring data persistence.

2.5 AI Response Evaluation

The AI-generated responses were evaluated based on:

- Relevance
- Clarity
- Structure
- Personalization

Result:
Responses were generally meaningful and context-aware, though further improvements can enhance accuracy.

3. Testing Summary Table

Table 4.1: Functional Testing and Validation Summary

Test Area	Expected Behaviour	Observed Result	Status
User/Profile Management	Store and retrieve user data	Successfully implemented	Pass
Session Handling	Create and switch sessions	Working correctly	Pass
Chat Functionality	Send and receive messages	Properly functioning	Pass
Data Persistence	Store and retrieve data	Successfully maintained	Pass
AI Response Generation	Generate relevant responses	Context-aware responses observed	Pass
UI/Usability	Easy navigation and readability	Improved interface	Pass

4. Observations and Discussion

4.1 System Performance

- The system performs efficiently for normal user interactions
- FastAPI ensures quick response handling
- SQLite provides fast data retrieval for small-scale usage

4.2 Strengths of the System

- Provides real-time career guidance
- Supports personalized responses
- Maintains conversation history
- Offers organized session management
- Simple and user-friendly interface

4.3 Limitations Observed

- AI responses may sometimes be generic without deeper context tuning
-
- Limited testing with a large number of users
- SQLite may not be suitable for large-scale deployment
- No real-time job data integration

4.4 Improvements Based on Testing

Based on testing results, the following improvements were identified:

- Enhance AI model accuracy
- Improve response personalization
- Add more interactive UI features
- Integrate external job and internship data sources

5. Insights from Development Phases

Testing results were aligned with development progress:

- Initial testing focused on basic functionality and workflow validation
- Implementation phase ensured database and session handling reliability
- Final phase improved UI, response quality, and overall usability

CHAPTER-5

CONCLUSIONS & FUTURE SCOPE

5.1 Conclusion

The **Generative AI Career Guidance Assistant** project successfully demonstrates the development of an intelligent, web-based system designed to support students in making informed career decisions. The project integrates modern technologies such as Generative AI, FastAPI, React, and SQLite to create a unified platform that provides personalized and continuous career guidance.

1. Summary of the Project

The system was developed to address key challenges faced by students, including:

- Lack of personalized career guidance
- Absence of continuous counselling support
- Difficulty in tracking previous discussions
- Unstructured career planning

To overcome these challenges, the project introduced a solution that combines:

- User profile management
- Session-based chat system
- Context-aware AI response generation
- Persistent data storage

2. Achievement of Objectives

The project successfully achieved its primary and specific objectives:

- Developed a profile-based system for personalized guidance
- Implemented a chat-based conversational interface
- Enabled multi-session interaction for organized discussions
- Integrated Generative AI for intelligent response generation
- Ensured data persistence using SQLite database
- Designed a user-friendly interface with improved usability

These achievements confirm that the system meets its intended goals.

3. Key Contributions of the Project

The project contributes to the field of career guidance systems by:

- Demonstrating the practical use of Generative AI in education
- Introducing a context-aware conversational model
- Providing a structured and continuous guidance platform
- Combining multiple technologies into a cohesive system

4. Learning Outcomes

The project provided valuable learning experiences, including:

- Understanding full-stack web development
- Working with backend frameworks like FastAPI
- Designing database systems using SQLite
- Integrating AI models into real-world applications
- Implementing session management and data persistence

5. Development Journey Reflection

The project development followed a structured approach:

- Initial phase focused on problem analysis and system design
- Implementation phase developed core functionalities and database integration
- Final phase improved user experience, personalization, and testing

This step-by-step progression ensured the successful completion of the project

6. Overall Evaluation

The system performs effectively as a prototype solution and demonstrates the potential of AI-driven career guidance systems. It provides:

- Interactive user experience
- Personalized recommendations
- Continuous and structured guidance

However, it is important to note that the system is currently at a prototype stage and requires further enhancements for real-world deployment.

5.2 Future Scope

Although the **Generative AI Career Guidance Assistant** has been successfully developed as a functional prototype, there are several opportunities for further enhancement and expansion. With continuous advancements in technology, the system can be improved to provide more accurate, scalable, and user-friendly career guidance.

1. Resume Analysis and Skill Evaluation

One of the most valuable future enhancements is the integration of a resume analysis module.

Proposed Features:

- Upload resume in PDF format
- Extract skills and experience

- Provide feedback on:
  - Content quality
  - Formatting
  - Missing skills

Benefit:
Helps students improve their resumes and align them with industry requirements.

2. Integration with Job and Internship Portals

The system can be enhanced by connecting it with real-time job and internship platforms.

Proposed Features:

- Fetch live job listings
- Recommend internships based on skills
- Provide application guidance

Benefit:
Bridges the gap between career guidance and actual job opportunities.

3. Advanced AI Model Integration

The current system uses a basic Generative AI setup. Future improvements can include:

- Fine-tuned AI models
- Domain-specific training
- Better context handling

Benefit:
Improves accuracy, relevance, and personalization of responses.

4. Voice-Based Interaction

To improve accessibility, the system can support voice input and output.

Features:

- Speech-to-text for user queries
- Text-to-speech for AI responses

Benefit:
Makes the system more interactive and accessible for all users.

5. Deployment and Cloud Integration

Currently, the system is a local prototype. Future work can include:

- Deployment on cloud platforms
- Multi-user support
- Secure authentication systems

Benefit:
Allows real-world usage and scalability.

6. Enhanced User Interface and Experience

Further UI improvements can be implemented:

- Better design and animations
- Dashboard for tracking progress
- Visual career roadmaps

Benefit:
Improves engagement and usability.

7. User Feedback and Evaluation System

A feedback mechanism can be added to:

- Rate AI responses
- Collect user suggestions
- Improve system performance

Benefit:
Helps in continuous improvement of the system.

8. Large-Scale Testing and Validation

Future work should include:

- Testing with a large number of users
- Performance evaluation under load
- Accuracy analysis of AI responses

Benefit:
Ensures system reliability and scalability.

9. Integration with Educational Platforms

The system can be connected with:

- Learning platforms (Coursera, Udemy, etc.)
- University portals

Benefit:
Provides direct learning resources along with guidance.

10. Insights from Development Progress

The future scope is aligned with observations made during development:

- Initial phases focused on building a basic working system
- Implementation phases highlighted the need for better integration and scalability
- Final phases emphasized improving usability and response quality

These insights guide future improvements.

APPENDIX

The appendix section provides supplementary information related to the project, including development logs, system modules, and additional implementation details that support the main content of the report.

Fortnight	Work Completed	Next Planned Work
1	Project scope defined, system workflow planned, frontend (React + Vite) setup initiated, backend (FastAPI) structure created, database schema planned	Develop profile module, implement database connectivity, design chat session logic
2	Backend development progressed, user and session management implemented, SQLite integration completed, chat interface developed with message rendering	Improve context handling, enhance UI, implement session switching, begin testing
3	System refined into a functional prototype, UI improvements added (dark theme, navigation), contextual AI responses enhanced, testing completed	Add advanced features like resume analysis, job integration, and deployment planning

These progress stages were derived from the project development reports and reflect a step-by-step incremental development approach

REFERENCES

Russell, S., & Norvig, P. (2021). Artificial intelligence: A modern approach (4th ed.). Pearson.

Goodfellow, I., Bengio, Y., & Courville, A. (2016). Deep learning. MIT Press.

Brown, T. B., Mann, B., Ryder, N., Subbiah, M., Kaplan, J., Dhariwal, P., Neelakantan, A., Shyam, P., Sastry, G., Askell, A., Agarwal, S., Herbert-Voss, A., Krueger, G., Henighan, T., Child, R., Ramesh, A., Ziegler, D. M., Wu, J., Winter, C., … Amodei, D. (2020). Language models are few-shot learners. Advances in Neural Information Processing Systems, 33, 1877–1901.

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., & Polosukhin, I. (2017). Attention is all you need. Advances in Neural Information Processing Systems, 5998–6008.

FastAPI. (n.d.). FastAPI documentation. https://fastapi.tiangolo.com/

React. (n.d.). React documentation. https://react.dev/

SQLite. (n.d.). SQLite documentation. https://www.sqlite.org/docs.html

Vite. (n.d.). Vite documentation. https://vitejs.dev/

Savickas, M. L. (2013). Career construction theory and practice. In S. D. Brown & R. W. Lent (Eds.), Career development and counseling (pp. 147–183). Wiley.

UNESCO. (n.d.). Guidance and counselling in higher education. UNESCO Publications.

Project Team. (2025). Fortnightly progress reports for Generative AI Career Guidance Assistant (Unpublished internal document).

OpenAI. (2023). GPT-4 technical report. https://arxiv.org/abs/2303.08774

Pressman, R. S., & Maxim, B. R. (2019). Software engineering: A practitioner's approach (9th ed.). McGraw-Hill.

Sommerville, I. (2016). Software engineering (10th ed.). Pearson.

Nielsen, J. (1994). Usability engineering. Morgan Kaufmann.

World Economic Forum. (2020). The future of jobs report 2020. https://www.weforum.org/reports

Citation Note

All sources used in this report have been cited in accordance with the guidelines of the American Psychological Association (APA), 7th edition. The reference list includes textbooks, peer-reviewed research articles, official software documentation, and internal project reports that informed the theoretical background, system design, and implementation of the Generative AI Career Guidance Assistant.

The cited literature provides foundational knowledge in the areas of **a**rtificial intelligence, machine learning, and career development, while the documentation sources support the technical aspects of the system, including frontend, backend, and database development. Internal project reports were used to document the development process and progression of the system.

Proper acknowledgment has been given to all sources to maintain academic integrity and avoid plagiarism. In-text citations and the reference list have been formatted consistently following APA (7th edition) standards.

Gantt Chart

Task	F1	F2	F3	F4	F5	F6	F7
Problem Study	X	X					
Literature Review	X	X	X				
Frontend Development		X	X	X			
Backend Development		X	X	X	X		
Database Integration			X	X	X		
AI Integration			X	X	X		
Testing & Refinement				X	X	X	
Documentation					X	X	X

Responsibility Matrix

RACI Matrix: Responsible, Accountable, Consulted, Informed

Task	Akshit	Deepak	Utsav
Project Planning	C	R	C
Literature Review	R	A	A
Frontend Design	R	C	A
Backend Development	R	R	C
Database Management	R	R	I
AI Integration	A	R	C
Testing	R	A	C